

Problem 4

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Problem 1

- (a) You are given that $f''(x) > 0$ for all x . Which of the following must be true about $f(x)$ on the region $0 \leq x \leq 2$?
- (i) There is a critical point between 0 and 2.
 - (ii) There is a local maximum, but not enough information is given to determine where.
 - (iii) f need not have a local maximum.
- (b) You are told that $f''(x) > 0$ for all x . Which of the following must be true about the graph of $y = f(x)$?
- (i) The graph is a straight line.
 - (ii) The graph crosses the x -axis at most once.
 - (iii) The graph is concave down.
 - (iv) The graph crosses the y -axis more than once.
 - (v) The graph is concave up.